IfG

TU Graz

WISSEN
TECHNIK
LEIDENSCHAFT

Lab programme 1: Routing

Navigation Systems

WS 2026/27

IfG

Lab 1: Routing
Aim



- Generate optimal routes
- Dijkstra algorithm
- Start node: place where one person of your group lives
- End nodes:
  - Basilika Mariatrost
  - Schloss Eggenberg
  - Shopping-Center Murpark

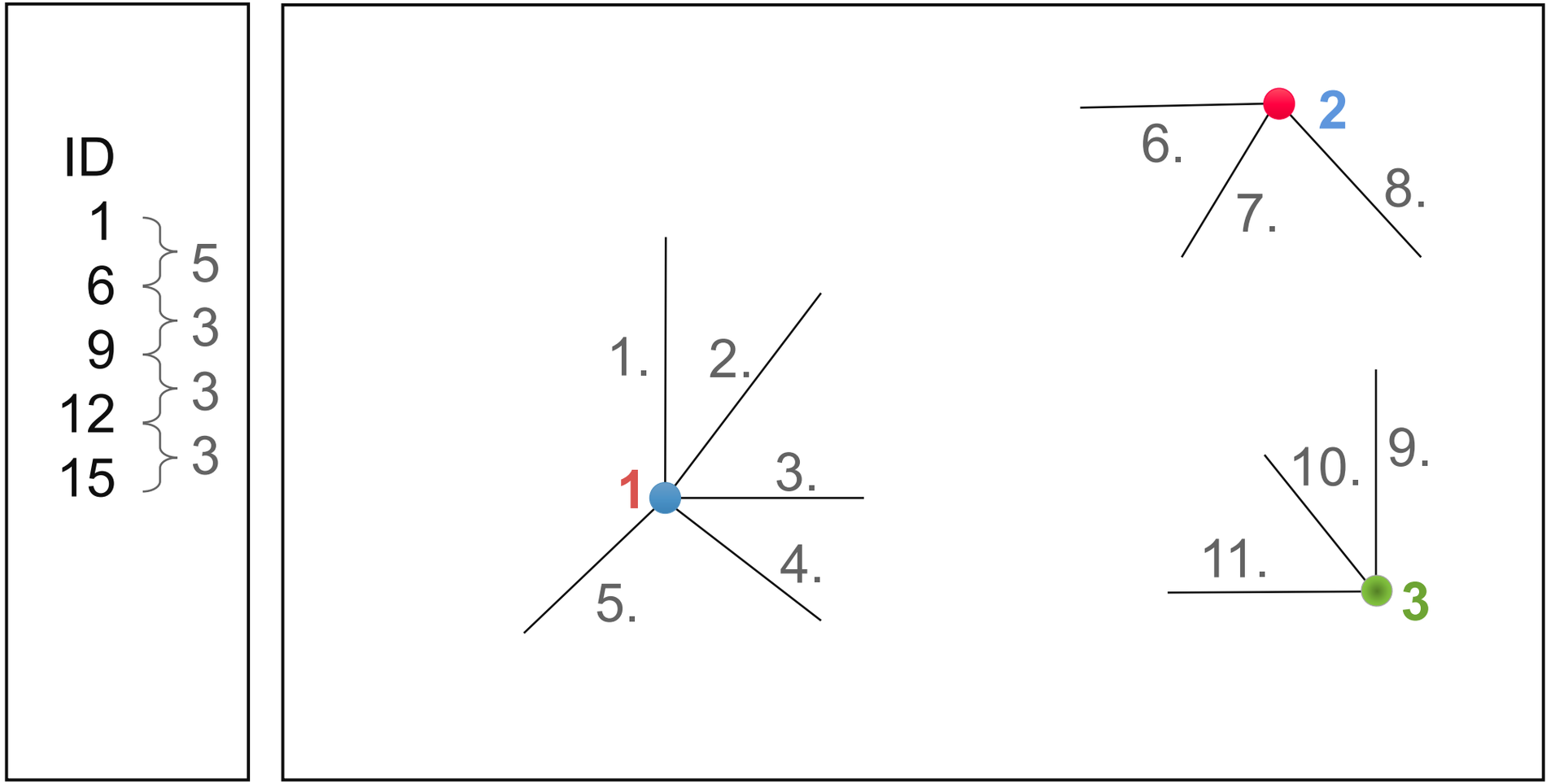




Lab 1: Routing
Given data

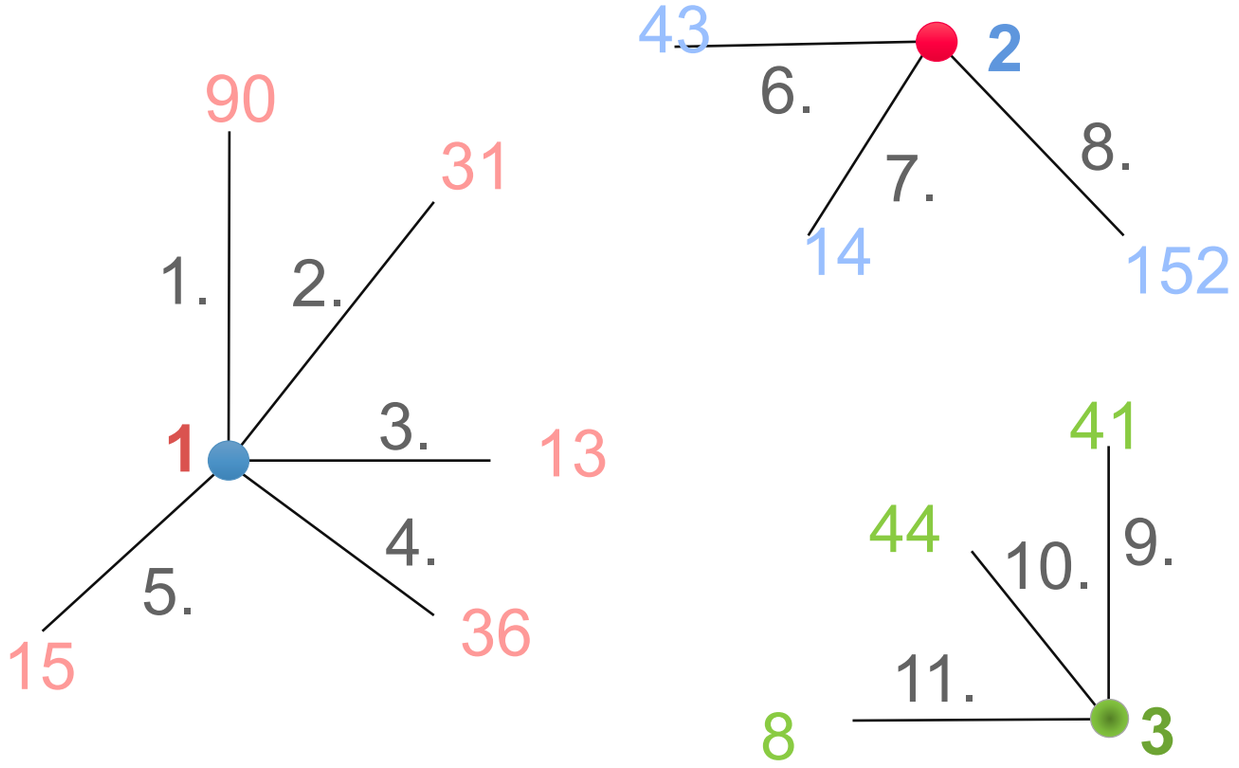
- nodepl.txt
  - Coordinates of nodes (φ and λ in degrees)
- nodelist.txt
  - Adjacency list of nodes
- arclist.txt
  - Adjacency list of arcs
  - Columns: time, distance, speed limit, clazz, flags

Adjacency list of nodes



Adjacency list of arcs

v_j	c_{ij} [m]
90	63.20
31	78.77
13	40.25
36	78.26
15	81.44
43	34.78
14	81.44
152	81.44
41	77.18
44	11.90
8	45.89



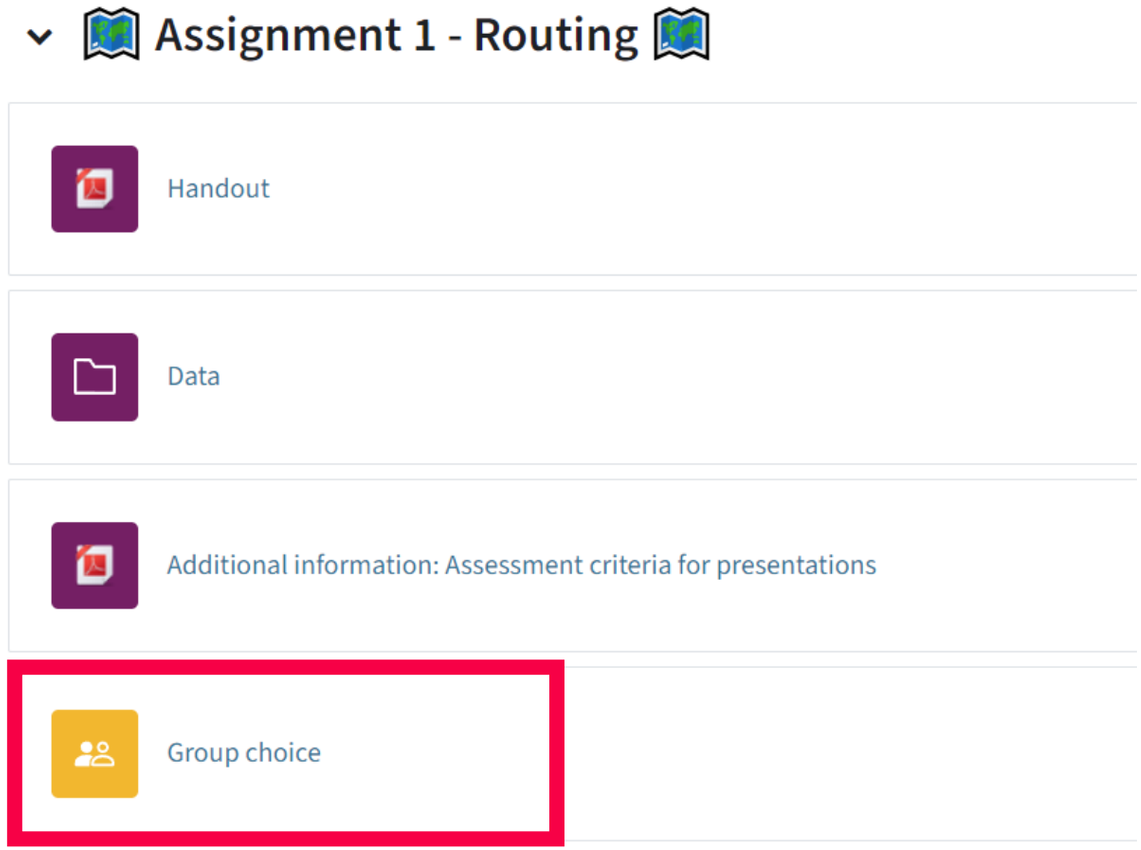
Lab 1: Routing
Tasks

- Cost functions
  - Time
  - Distance

- Additionally, choose between:
- 2.1 Self-developed cost-function for Dijkstra’s algorithm
- 2.2 Heuristic A* algorithm

Group choice

- Groups of two
- Sign up for a group in the TeachCenter
- Interdisciplinary collaboration is strongly encouraged

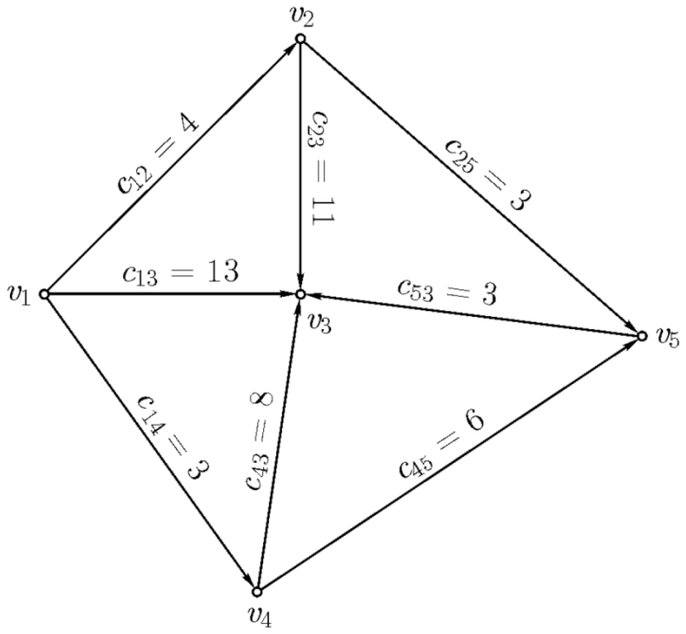


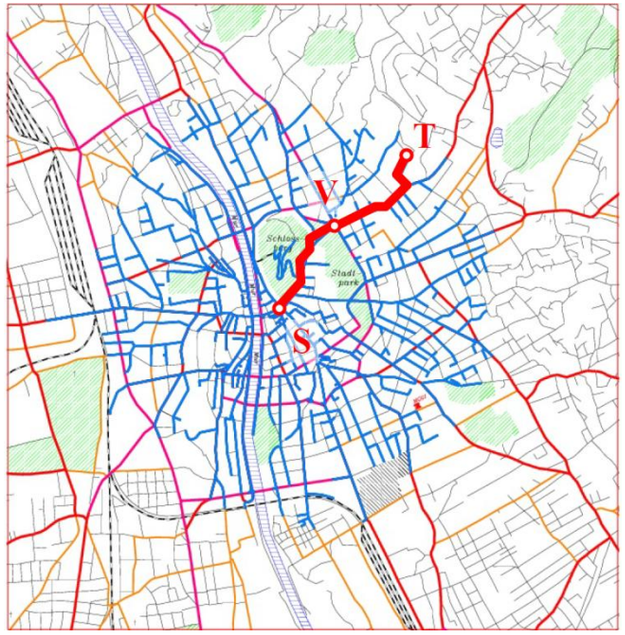
Lab 1: Routing
Deliverables

- Source code
- Presentation
  - Visualization of optimal routes (time & distance)
  - Evaluation of routes
  - Self-made cost function *or* A*-algorithm
  - 5 minutes video with audio commentary

Dijkstra algorithm

- Check with small amount of data
  - Example from lecture





Results and visualization

- prelist.txt
  - Store your predecessors
  - Necessary to visualize route
- Visualization
  - Be aware of aspect ratio
  - Label your graph
  - Use appropriate line width



Evaluation

- Lab due: 22nd November, 11:59 p.m.
- Source code of developed software
- Short presentation
  - Presentation: Video with audio commentary (.mp4)
  - 5 minutes

Best presentation award



2 best presentations will be rewarded with a small prize and will be put on TUBE